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UGANDA ADVANCED CERTIFICATE OF EDUCATION (UACE)

END OF FIRST TERM EXAMINATIONS 2026 — SET 4

S.6 P425/1 PURE MATHEMATICS (Paper 1)

Time Allowed: **3 HOURS**

STUDENT NAME:

.....

PERSONAL NUMBER: ..... **SIGNATURE:** .....

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO BY THE INVIGILATOR

INSTRUCTIONS TO CANDIDATES:

1. Do not write anything in this paper; any rough work should be written in the response booklet(s) provided and crossed out.
2. The time allowed for this paper is strictly **3 Hours** and no additional time shall be granted.
3. This paper consists of **three** sections; **A, B** and **C** with a total of **6 items**.
4. Section A has 2 items, attempt **any one** item.
5. Section B has **One COMPULSORY** item.
6. Section C has **Three** items, attempt any **two** of your choice.
7. **FOUR ITEMS** should be attempted in total. Any additional item(s) attempted will not be marked.
8. Begin each item on a fresh page and clearly indicate them in the response booklet provided.
9. Tidy handwriting, meaningful mathematical judgement and conclusion may increase your chances of excelling.
10. Silent non-programmable calculators and logarithmic tables may be used.

FOR SCORER'S USE ONLY

SECTION	A		B	C			TOTAL
ITEM	1	2	3	4	5	6	
SCORE							
INITIAL							

SECTION A

Attempt any One item in this Section

ITEM 1

Dr. Tumwine is the Chief Actuary at a leading insurance company in Kampala. He uses advanced algebraic techniques daily: approximating risk indices using binomial expansions, proving financial formulae by induction, and modelling claim dispersion patterns using complex number loci.

A risk adjustment index used in the company's pricing model is computed from the expression $(8 + 3x)^{-\frac{1}{3}}$. Dr. Tumwine needs to expand this expression in ascending powers of x up to and including the term in x^3, stating the range of values of x for which the expansion is valid. He then uses this expansion with a suitable value of x to find an approximation for $\sqrt[3]{(1/9)}$, giving his answer to 4 significant figures.

For an actuarial proof used in the company's financial reserve documentation, Dr. Tumwine must establish by mathematical induction that for all positive integers n:

$$\Sigma\, r(r + 1) = \tfrac{1}{3}\, n(n + 1)(n + 2) \quad \text{for all positive integers } n.$$

The company's claims dispersion model maps complex numbers $z = x + iy$ onto a risk plane. A claim region is defined by the locus of a complex number z such that the argument of $(z - 2) / (z - 2i)$ equals $\pi/4$. Dr. Tumwine must find the Cartesian equation of this locus and describe it geometrically.

His major objectives are to produce the risk index approximation, prove the actuarial reserve formula by induction, and determine the claim dispersion locus.

Task:

Help Dr. Tumwine to:

a) (i) Expand $(8 + 3x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the x^3 term, stating the values of x for which the expansion is valid.

(ii) Hence, by choosing a suitable value of x, find an approximation to $\sqrt[3]{(1/9)}$ correct to 4 significant figures.

b) Prove by mathematical induction that $\Sigma_{r=1}^{n}\, r(r + 1) = \tfrac{1}{3}\, n(n + 1)(n + 2)$ for all positive integers n.

c) Find the Cartesian equation of the locus of a complex number z such that $\arg\wedge((z - 2)/(z - 2i)) = \pi/4$. Describe the locus geometrically.

ITEM 2

Ms. Kemigisha is a cryptographic systems engineer at the National Information Technology Authority of Uganda (NITA-U). She uses algebraic decomposition, series analysis, and roots-of-unity arguments to design, verify and break down encryption algorithms.

A signal-processing filter function in the encryption algorithm is given by:

$$f(x) = (5x^2 + 3x + 1) / [(x - 1)(x^2 + x + 1)]$$

Ms. Kemigisha must express f(x) in partial fractions. She then uses the partial fraction decomposition to evaluate the definite integral of f(x) from x = 2 to x = 4.

The algorithm's key-scheduling sequence is defined by the recurrence relation $u_{n+1} = 3u_n + 2$, with $u_1 = 1$. Ms. Kemigisha needs to find a closed-form expression for u_n and hence compute the sum $S_n = \Sigma_{r=1}^{n} u_r$ of the first n terms of the sequence.

The final stage of the encryption key uses all three cube roots of unity, ω, where $\omega^3 = 1$. Ms. Kemigisha must show that $1 + \omega + \omega^2 = 0$ and use this result to prove that if $p + q\omega + r\omega^2 = 0$, where p, q, r are real, then p = q = r. She then applies this result to find the values of p, q and r given that $p + 2q\omega + 3r\omega^2 = 0$.

Her major objectives are to decompose the filter function and evaluate the integral, determine the closed-form key-scheduling formula and series sum, and establish the cube roots of unity properties for the encryption key.

Task:

Help Ms. Kemigisha to:

a) (i) Express $f(x) = (5x^2 + 3x + 1) / [(x - 1)(x^2 + x + 1)]$ in partial fractions.

(ii) Hence evaluate $\int_2^4 f(x)\,dx$, giving your answer in the form a + b ln c where a, b, c are rational.

b) (i) Find a closed-form expression for u_n given that $u_{n+1} = 3u_n + 2$, $u_1 = 1$.

(ii) Hence find $S_n = \Sigma_{r=1}^{n} u_r$ in simplified form.

c) Given that ω is a complex cube root of unity, show that $1 + \omega + \omega^2 = 0$. Hence prove that if $p + q\omega + r\omega^2 = 0$ for real p, q, r, then p = q = r. Find p, q, r given $p + 2q\omega + 3r\omega^2 = 0$.

SECTION B

This section has ONE Compulsory Item

ITEM 3

Eng. Nakayima is a senior design engineer at the Uganda National Bureau of Standards (UNBS). She applies advanced geometry in testing the curvature of manufactured components, verifying engineering drawings, and analysing spatial structures in quality control assessments.

A component edge profile is traced by the parametric curve $x = t - \sin t$, $y = 1 - \cos t$. Eng. Nakayima needs to find the gradient of the curve at parameter $t = \pi/3$, determine the equation of the tangent and normal to the curve at this point, and eliminate the parameter to express the relationship between y and x near this point using a small-angle approximation ($t \approx 0$) to find the Cartesian equation valid for small t.

In testing a circular component, a circular cross-section has equation $x^2 + y^2 - 6x - 4y + 4 = 0$. An external inspection probe is positioned at point T(8, 1) and must be tangent to the circle. Eng. Nakayima needs to find the length of the tangent from T to the circle, the equations of both tangent lines from T to the circle, and the equation of the chord of contact.

A spatial quality frame has three edge vectors: $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, and $\mathbf{c} = 5\mathbf{i} + \lambda\mathbf{j} + \mu\mathbf{k}$. Eng. Nakayima must find the values of λ and μ for which vectors **a**, **b** and **c** are coplanar, and then find the unit vector perpendicular to both **a** and **b**.

Her major objectives are to analyse the parametric component profile, determine the tangent properties of the circular cross-section, and establish the spatial coplanarity conditions and perpendicular direction.

Task:

Help Eng. Nakayima to:

a) For the parametric curve $x = t - \sin t$, $y = 1 - \cos t$:

(i) Find the gradient of the curve at $t = \pi/3$.

(ii) Find the equations of the tangent and normal to the curve at $t = \pi/3$.

(iii)For small values of t, show that $y \approx \frac{1}{2}t^2$, and find the corresponding approximation for x. Hence find the approximate Cartesian equation near the origin.

b) For the circle $x^2 + y^2 - 6x - 4y + 4 = 0$ and external point T(8, 1):

(i) Find the length of the tangent from T to the circle.

(ii) Find the equation of the chord of contact from T.

c) Find the values of λ and μ such that **a**, **b** and **c** are coplanar. Hence find a unit vector perpendicular to both **a** and **b**.

SECTION C

Attempt any Two items from this Section

ITEM 4

Dr. Nansamba is a biomedical engineer at Mulago National Referral Hospital. She uses advanced calculus to model the cross-sectional profiles of artery implants, analyse oscillatory blood-flow signals, and compute the volumetric properties of vascular stent designs.

The cross-sectional boundary of an artery implant prototype is modelled by the implicit curve:

$$x^3 + y^3 - 3xy = 0 \quad \text{(the Folium of Descartes, scaled)}$$

Dr. Nansamba needs to find dy/dx by implicit differentiation, locate all points on the curve where the tangent is horizontal, and verify the nature of these points.

An oscillatory blood-flow signal produces a force function proportional to $e^x \cos x$. To compute the total impulse of this signal over a cycle, Dr. Nansamba needs to find the indefinite integral $\int e^x \cos x \, dx$, showing all working clearly.

A vascular stent has a cross-section bounded by the curve $y = \sqrt{(\sin x)}$ for $0 \leq x \leq \pi$. The stent is manufactured by rotating this region through 2π radians about the x-axis. Dr. Nansamba must compute the exact volume of the solid of revolution generated.

Her major objectives are to analyse the implant boundary curve, evaluate the blood-flow impulse integral, and determine the volumetric specification of the stent.

Task:

Help Dr. Nansamba to:

a) For the curve $x^3 + y^3 - 3xy = 0$:

(i) Find dy/dx by implicit differentiation.

(ii) Find the coordinates of the point(s) on the curve where the tangent is horizontal. (Do not include the origin.)

b) Using integration by parts twice, find $\int e^x \cos x \, dx$.

c) Find the volume of the solid formed when the region bounded by $y = \sqrt{(\sin x)}$, $0 \leq x \leq \pi$, is rotated through 2π radians about the x-axis. Give your answer as an exact multiple of π.

ITEM 5

Prof. Mutyaba is a professor of Applied Mathematics at Kyambogo University. He is developing a research paper on integral reduction techniques, first-order system modelling, and series approximations for computational engineering applications.

In his research on integral reduction, Prof. Mutyaba defines the integral:

$$I_n = \int_0^{(\pi/2)} x^n \cos x \, dx$$

He needs to show that I_n satisfies the reduction formula $I_n = -n(n-1)I_{n-2} + (n\pi/2)^n$, and use this formula with I_0 and I_1 to evaluate I_4 exactly.

A dynamic heat-flow system is governed by the first-order differential equation:

$$dy/dx + (2/x)\,y = x^2 + 1, \quad x > 0$$

Prof. Mutyaba must solve this equation completely using an integrating factor, applying the initial condition $y = 3$ when $x = 1$, to find the particular solution.

For a computational approximation library, Prof. Mutyaba needs to find the Maclaurin series expansion of $f(x) = \ln(\cos x)$ up to and including the term in x^4. He then uses this series to find an approximation for $\int_0^{0.1} \ln(\cos x)\, dx$, giving the answer to 6 decimal places.

His major objectives are to establish the reduction formula and evaluate I_4, solve the heat-flow differential equation, and produce the Maclaurin series approximation for the integral.

Task:

Help Prof. Mutyaba to:

a) (i) Show that $I_n = \int_0^{(\pi/2)} x^n \cos x \, dx$ satisfies the reduction formula: $I_n = (n\pi/2)^{n-1} - n(n-1)I_{n-2}$.

(ii) Starting from $I_0 = 1$ and $I_1 = \pi/2 - 1$, evaluate I_4 exactly.

b) Using an integrating factor, solve $dy/dx + (2/x)\,y = x^2 + 1$ given $y = 3$ when $x = 1$.

c) (i) Find the Maclaurin series of $\ln(\cos x)$ up to and including the term in x^4.

(ii) Hence find an approximation for $\int_0^{0.1} \ln(\cos x)\, dx$, correct to 6 decimal places.

ITEM 6

Mr. Opolot is a senior process engineer at the Uganda Sugar Corporation in Lugazi. He models the rate at which sugar syrup fills storage tanks, analyses average concentration levels in mixing vessels, and solves second-order differential equations governing the vibration of machinery in the factory.

A conical storage tank has its vertex pointing downwards. The half-angle at the vertex is 30° (so that the radius r and depth h of liquid satisfy $r = h/\sqrt{3}$). Syrup is pumped into the tank at a constant rate of 2 m^3 per minute. When the depth of syrup is 3 m, Mr. Opolot needs to find the rate at which the depth is increasing and the rate at which the surface area of the syrup (a circle of radius r) is increasing at that instant.

The concentration of a sugar solution (in g/L) in a mixing vessel at time t seconds is modelled by $C(t) = (2t + 1)e^{-t}/2$ for $0 \leq t \leq 4$. Mr. Opolot needs to find the mean concentration over this time interval.

The vertical displacement y (in mm) of a vibrating factory component satisfies the second-order differential equation:

$$d^2y/dx^2 - 5(dy/dx) + 6y = 2e^{3x}$$

Mr. Opolot must find the general solution of this equation by finding the complementary function and a particular integral, and then apply the initial conditions $y = 0$ and $dy/dx = 1$ when $x = 0$ to find the particular solution.

His major objectives are to find the rates of change of the syrup depth and surface area, determine the mean concentration in the mixing vessel, and find the complete vibration displacement equation.

Task:

Help Mr. Opolot to:

a) For the conical tank ($r = h \tan 30° = h/\sqrt{3}$), with $dV/dt = 2\ m^3/min$:

(i) Find dh/dt when $h = 3$ m.

(ii) Find the rate at which the surface area of the syrup ($A = \pi r^2$) is increasing when $h = 3$ m.

b) Find the mean value of $C(t) = (2t + 1)e^{-t}$ over the interval $0 \leq t \leq 4$.

c) Find the general solution of $d^2y/dx^2 - 5(dy/dx) + 6y = 2e^{3x}$. Hence find the particular solution satisfying $y = 0$ and $dy/dx = 1$ when $x = 0$.